

Multimodal RAG Strategy

Indexing channels

Five indexes are built from the approved inventory in a single pass. Three carry semantic content; one carries identifiers; one carries the layer ontology.

Channel	Backend	Content	When it wins
Text chunks	ChromaDB + FTS5	prose page blocks from text-rich documents (A4)	functional questions, prose paraphrases
Captions	ChromaDB	per-asset <code>caption + vision_prompt</code>	semantic diagram retrieval
Images	ChromaDB	per-asset crop PNG	visual-similarity recall
Identifiers	FTS5	deduped <code>display_name + caption</code> per asset	exact part IDs, drawing numbers
Layer descriptors	ChromaDB	one vector per layer (<code>name + description + keywords</code>)	not queried directly — used to re-score vector hits by layer affinity

Substrate ordering

- **Captions are the primary diagram-retrieval channel.** They are inspectable and editable.
- **Image embeddings are secondary.** Opaque, expensive to debug — used as visual recall, not as the primary signal.
- **Identifier tokens are deduped to a token bag.** BM25 doesn't over-weight tokens that appear in both `display_name` and `caption` .

Caption language policy

- Captions are always written in **English**, regardless of the source language.
- The text embedder is multilingual. Page-block vectors in the `text_chunks` collection are built from the **English translation** of each block (computed once at build time, also stored in the FTS5 `text_en` column); English-source blocks fall through to source text directly. So an English query reliably hits any indexed block via the vector channel, regardless of the source PDF's language.
- For queries in the source language itself (e.g. a German query against a German manual), the FTS5 keyword channel still indexes the source-language text in its `text` column, preserving same-language keyword recall.

Embedder choices

- **Text and captions** — BAAI/bge-m3 (1024-d, multilingual, 8k context).
- **Images** — ViT-B-16-SigLIP (768-d, joint text-image space, L2-normalized).
- **Re-Ranker** — BAAI/bge-reranker-v2-m3 (local Cross-Encoder re-scorer).

The dual-encoder design of the image embedder is load-bearing: the same model embeds crop PNGs at build time and text queries at retrieval time, so visual-style questions reach the image collection directly. Both the crop image vectors and the text query vectors are L2-normalized to unit length. This mathematical normalization is essential to ensure precise, un-skewed cosine similarity matching: without scaling both vectors to unit length, raw un-normalized vector magnitudes would skew the scoring inside the vector database, corrupting similarity distances and degrading retrieval precision.

Large-format readability — two-pass cropping

Large-format drawings (A2, A1, A0) cannot be rendered at legible DPI within the 40 Mpx pixel cap. Both cropping agents (the region-proposal agent and the re-cropping agent) work around it with a two-pass strategy.

Pass 1 — Grid overview ("the map")

- Renders the full page at 120 DPI with a normalized coordinate grid overlay (alphanumeric cell labels starting at `A1`, axis labels `0.0` – `1.0`). Grid density adapts to page size so cells stay fine enough to disambiguate small features on the largest formats.

- Text is blurry but the layout is clear.
- The agent identifies what's on the page and reads off normalized coordinates from the grid cells.

Pass 2 — High-resolution crop ("the magnifying glass")

- Renders a chosen bbox at up to 600 DPI as PNG.
- Callout numbers, dimension text, and fine annotations become legible.
- The agent verifies labels, traces leader endpoints, and commits the final crop.

DPI budget

Format	Grid overview	High-res crop	Notes
A4	120	300	a single render often fits within the pixel cap
A3	120	300–400	tight crops for tables and callouts
A2	120	300–400	grid essential for spatial orientation
A1	120	300–400	multiple crops per page typical
A0 / oversized	120	300–400	pixel cap may force lower DPI on wide crops (rule A.13)

Vision-degradation policy

If a label is illegible at the rendered DPI:

1. Escalate — call the high-res crop with a higher DPI or tighter bbox.
2. If still illegible at the max DPI, cite as `[doc_id, p.N – label illegible]` and note it in the region's description.
3. Never guess a label value.

Automated quality gates before HITL

Two automated steps run before the human takes over.

Crop quality gate

- **Verifier** (Gemini 3.5 Flash) inspects each crop against the image-quality rules tagged `<!--quality-->` in the system rules manifest.
- The verifier model is **pinned smaller than the cropping agent** on purpose — a weaker judge over a stronger producer prevents rationalization (independence by asymmetry).
- Failures feed the **re-crop agent**, which adjusts the bbox and re-renders.
- The verify ↔ re-crop loop runs up to **3 rounds**. Remaining failures surface for HITL review.

Layer assignment

- After URNs are minted, the vision model sees `crop image + caption + vision_prompt + ontology` and picks one layer per asset.
- The URN is unchanged — layer is a free field on the asset, not encoded in the identifier.
- The human approves or overrides at HITL Gate 4.

On-demand integrity check

The automated integrity check runs fast local checks: corpus hash consistency, crop file existence, region link validity, flag sanity. It also runs internally as the first step of the catalog export process to refuse stale data before machine catalog emission.

Hybrid retrieval

A query fans out across all channels independently. Vector hits are re-scored against layer affinity, and Reciprocal Rank Fusion combines the channels before Stage 2 Re-Ranking.

Routing

A single regex tags each query as `LOOKUP` (contains an asset-ID token like `S-304`) or `SEMANTIC` (everything else). Each class carries its own per-channel weights. A visual-dominant retrieval preset swaps in an image-dominant weight matrix for visual-style sessions.

Class	Trigger	Channel weights
<code>LOOKUP</code>	query contains an asset-ID-like token	text 0.45, captions 0.25, id 0.20, image 0.10
<code>SEMANTIC</code>	everything else	image 0.25, text 0.40, captions 0.30, id 0.05
Visual-dominant preset	user opt-in	image 0.60, text 0.05, captions 0.30, id 0.05

If `LOOKUP` returns too few hits, the system re-fuses with `SEMANTIC` weights automatically.

Fan-out

Every query reaches every channel independently — channels are never gated on each other.

- **Vector** — encode the query once with the text embedder, hit the text-chunks and captions collections.
- **Image** — encode the **same query text** with the image model's text encoder, L2-normalize to unit length, and hit the images collection.
- **Keyword** — sanitize the query into a safe FTS5 expression, hit both the prose table and the identifier table.

Cross-language keyword matching

The prose FTS5 table carries two columns per block: the source-language text and an English translation produced once at build time. A query in either language matches without extra logic.

Section-aware translation

- Each block is sent to the translator together with the name of the section heading that encloses it.

- The heading is detected automatically at document seed time and attached to the document.
- CAD-shaped documents are skipped — they have no TOC.
- Effect: technical terms whose meaning depends on the section resolve correctly (e.g. the same Slovenian word means *level* in a hydraulic section and *grade* in a safety section).

Layer-aware blend

Before fusion, every vector hit is re-scored as:

$$\text{score}' = \alpha \cdot \text{sim_layer} + (1 - \alpha) \cdot \text{sim_passage}$$

- `sim_layer` is the cosine similarity between the query and the hit's layer descriptor.
- The layer descriptor embeds `name + description + keywords` — three complementary signals (anchor, meaning, keyword hooks).
- Default `$\alpha = 0.3$` , tunable per query (via the layer weight setting); `$\alpha = 0$` disables the blend.
- Layer-flavoured queries pull the matching layer up; layer-agnostic queries contribute near-zero layer weight and rank by content alone.
- Keyword hits pass through unchanged.

The passage vectors stay **clean** — no layer prefix is mixed into the embedded text, so passage similarity reflects content meaning undiluted. The layer signal lives entirely in the separate descriptor collection and the α blend.

Fusion

Reciprocal Rank Fusion with `k = 60` combines the channels using the routed weights:

$$\text{score} = \sum \text{weight_channel} / (60 + \text{rank_channel})$$

Top candidates from this Stage 1 fusion are passed directly to Stage 2 Re-Ranking.

Two-stage retrieval — Cross-Encoder Re-Ranking

Once Stage 1 (RRF fusion) combines the keyword, vector, and visual channels to yield the top **30 candidate hits**, the system executes an automated **Stage 2 Re-Ranking** pass before handing the context blocks to the Q&A synthesis model.

- **The Model:** `BAAI/bge-reranker-v2-m3` (local Cross-Encoder).
- **How it works:** The re-ranker takes the query and the candidate passages, concatenates them, and runs full transformer self-attention over the combined sequence. This scores direct query-passage token interactions to output highly precise matching probabilities.
- **Re-Scoring:** Every candidate is assigned its new Cross-Encoder score, and candidates are re-sorted descending. Standard retrieval top-K elements are sliced from this re-sorted list, concentrating citable truth and filtering out semantic noise with perfect accuracy.